

MODULE DESCRIPTION FORM

نموذج وصف المادة الدراسية

Module Information				
معلومات المادة الدراسية				
Module Title	Modern Programming Language		Module Delivery	
Module Type	Elective		☒ Theory ☐ Lecture ☒ Lab ☐ Tutorial ☐ Practical ☐ Seminar	
Module Code	COMP214			
ECTS Credits	5			
SWL (hr/sem)	125			
Module Level	3	Semester of Delivery		5
Administering Department	CS	College	CS and Mathematics	
Module Leader	Prof. Dr. Ahmed Hussein Aliwy		e-mail	ahmedh.almajidy@uokufa.edu.iq
Module Leader's Acad. Title		Module Leader's Qualification	Ph.D.	
Module Tutor	Prof. Dr. Ahmed Hussein Aliwy		e-mail	ahmedh.almajidy@uokufa.edu.iq
Peer Reviewer Name	Name	e-mail	E-mail	
Scientific Committee Approval Date	01/06/2023	Version Number	1.0	

Relation with other Modules			
العلاقة مع المواد الدراسية الأخرى			
Prerequisite module	Advanced Programming Language	Semester	2
Co-requisites module	---	Semester	-

Module Aims, Learning Outcomes and Indicative Contents أهداف المادة الدراسية ونتائج التعلم والمحتويات الإرشادية	
Module Objectives أهداف المادة الدراسية	Computer programming skills are now becoming part of basic education as these skills are increasingly of vital importance for future job and career prospects. The Python programming language is one of the most popular programming languages worldwide. The course shows how to use the free open-source Python to write basic programs and high level applications using concepts such as Class, BIF of Python, functions, variables, If Else statements, For loops, While loops, iterative and recursive programs and algorithms such as the Insertion Sort algorithm, and dealing with file. This course will be of great interest to all learners who would like to gain a thorough knowledge and understanding of the basic components of computer programming using the Python language – and might be a gentle introduction to programming for those who think they might have a longer term interest in the subject area. This module aims to:
Module Learning Outcomes مخرجات التعلم للمادة الدراسية	1. Knowledge and understanding A student completing this module should be able to: a. Identify object-oriented design concepts, teach the knowledge and skills. b. Recognize and use several core data structures: Lists, Dictionaries, Tuples, and Strings. c. Extends students' proficiency in programming in object-oriented and procedural environment. d. Recognize basic programming paradigms including object-oriented, imperative, and functional programming . 2. Cognitive skills (thinking and analysis). a. Acquire a full Object Oriented Thinking. b. Be able to design small object oriented programs which meet requirements expressed in English. c. Be able to develop programs in python programming language. 3. Practical skills. a. Use object-oriented programming techniques to design and implement a clear, well-structured Python program b. Apply Python Exception Handling model to develop robust programs and learn how to use it. c. Write and debug Python programs which make use of the fundamental control d. Use API libraries for python e. Recognize basic programming paradigms including object-oriented, imperative, and functional programming
Indicative Contents المحتويات الإرشادية	The course focuses on problem solving strategies and the use of programing language to make applications.

Learning and Teaching Strategies استراتيجيات التعلم والتعليم	
Strategies	The main strategy that will be adopted in delivering this module is to encourage

	students' participation in the exercises, while at the same time refining and expanding their critical thinking skills. This will be achieved through classes, interactive tutorials, lab hours and by considering types of some small projects that are interesting to the students.
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Student Workload (SWL)			
الحمل الدراسي للطالب محسوب لـ ١٥ اسبوعا			
Structured SWL (h/sem) الحمل الدراسي المنتظم للطالب خلال الفصل	80	Structured SWL (h/w) الحمل الدراسي المنتظم للطالب أسبوعيا	5
Unstructured SWL (h/sem) الحمل الدراسي غير المنتظم للطالب خلال الفصل	45	Unstructured SWL (h/w) الحمل الدراسي غير المنتظم للطالب أسبوعيا	3
Total SWL (h/sem) الحمل الدراسي الكلي للطالب خلال الفصل	125		

Module Evaluation					
تقييم المادة الدراسية					
		Time/Number	Weight (Marks)	Week Due	Relevant Learning Outcome
Formative assessment	Quizzes	2	10% (10)	5 and 10	All
	Assignments	4	20% (10)	4,7,10 and 12	All
	Projects / Lab.				All
	Report	1	10% (10)	13	All
Summative assessment	Midterm Exam	2hr	10% (10)	11	All
	Final Exam	3hr	50% (50)	16	All
Total assessment			100% (100 Marks)		

Delivery Plan (Weekly Syllabus)	
المنهاج الاسبوعي النظري	
	Material Covered
Week 1	Python history, Characteristics ,Versions , operators types and Variables
Week 2	Python basics, importing module , Mutable and Immutable Data Type
Week 3	Elementary Control Structures
Week 4	Additional Control Structures

Week 5	Functions and Modules
Week 6	Global and local variables, Lambda expression
Week 7+8	Dealing with string, Lists, Tuples and Dictionaries
Week 9+10	Input, Output, and Files
Week 11	Midterm exam
Week 12 +13	Classes and Deeper Understanding of the Management of Objects
Week 14	Scripts for publishing on the Web
Week 15	Advance topic
Week 16	Final Exam

Delivery Plan (Weekly Lab. Syllabus) المناهج الاسبوعي للمختبر	
	Material Covered
Week 1	Introduction to Python environment and installing , Interactive mode and Script Mode
Week 2	operators types, Built-in functions and None Built-in functions
Week 3	False and True in python, if...elif ...else statement
Week 4	range() statement , for, while, continue and break
Week 5+6	Default argument in functions, Variable-length arguments in functions
Week 7	Global and local variables, Lambda expression
Week 8	string, Lists, Tuples, Dictionaries and their functions
Week 9+10	Files path, os module, reading and writing to text files
Week 11	Midterm exam
Week 12+13	Class , Inheritance and overriding
Week 14	Scripts for publishing on the Web
Week 15	Advance topic
Week 16	Final Lab Exam

Learning and Teaching Resources مصادر التعلم والتدريس		
	Text	Available in the Library?
Required Texts	Gowrishankar S , Veena A (2018) “Introduction to Python Programming”, Chapman and Hall/CRC; 1 edition .	No

	Richard L. Halterman "Fundamentals of Python Programming" http://python.cs.southern.edu/pythonbook/pythonbook.pdf	
Recommended Texts	- Mark Lutz and David Ascher, "Learning Python", Beijing: O'Reilly, 2004, 2nd ed.. - Dusty Phillip, "Python 3 Object Oriented Programming", Packt, July 2010. (ISBN : 1849511268 ISBN 13 : 978-1-849511-26-1)	No
Websites	http://www.python.org/	

Grading Scheme مخطط الدرجات				
Group	Grade	التقدير	Marks %	Definition
Success Group (50 - 100)	A – Excellent	امتياز	90 - 100	Outstanding Performance
	B - Very Good	جيد جدا	80 - 89	Above average with some errors
	C – Good	جيد	70 - 79	Sound work with notable errors
	D - Satisfactory	متوسط	60 - 69	Fair but with major shortcomings
	E - Sufficient	مقبول	50 - 59	Work meets minimum criteria
Fail Group (0 – 49)	FX – Fail	راسب (قيد المعالجة)	(45-49)	More work required but credit awarded
	F – Fail	راسب	(0-44)	Considerable amount of work required
Note: Marks Decimal places above or below 0.5 will be rounded to the higher or lower full mark (for example a mark of 54.5 will be rounded to 55, whereas a mark of 54.4 will be rounded to 54. The University has a policy NOT to condone "near-pass fails" so the only adjustment to marks awarded by the original marker(s) will be the automatic rounding outlined above.				